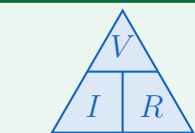
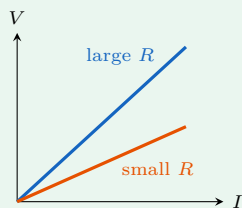


1. OHM'S LAW (MUST KNOW)



Cover the one you want!



$$V = IR \quad I = \frac{V}{R} \quad R = \frac{V}{I}$$

V volt (V) I ampere (A)
 R ohm (Ω) $Q = It$ coulomb (C)

$V-I$ graph: straight line through origin = **ohmic** conductor. **Gradient** = R . Steeper line = bigger resistance.

2. RESISTANCE OF A WIRE

$$R = \frac{\rho L}{A}$$

- ρ = resistivity ($\Omega \text{ m}$), L = length (m), A = cross-section area (m^2)
- Longer** wire $\Rightarrow$ **more** R **Thicker** wire $\Rightarrow$ **less** R
- Metals: R rises as temperature rises

3. POWER & ENERGY

$$P = IV \quad P = I^2 R \quad P = \frac{V^2}{R}$$

$$E = Pt = IVt \quad P = \frac{E}{t}$$

P : watt (W) = J/s

E : joule (J)

t : seconds (s)

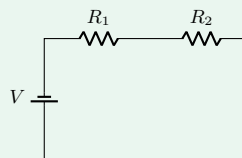
1 kWh = 3.6×10^6 J

Energy in kWh: $E = P(\text{kW}) \times t(\text{hours})$

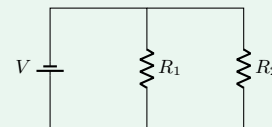
EXAM TIP

Pick the power formula from what is given: **I and $V \rightarrow IV$; I and $R \rightarrow I^2 R$; V and $R \rightarrow V^2/R$. Write units in every step.**

4. SERIES vs PARALLEL



SERIES (one path)



PARALLEL (branches)

	Series	Parallel
Current	same everywhere	splits: $I = I_1 + I_2$
Voltage	splits: $V = V_1 + V_2$	same on each branch
Total R	$R_T = R_1 + R_2 + \dots$	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$
Effect	R_T is bigger than any one	R_T is smaller than any one
One breaks	all stop	others keep working

Two resistors in parallel: $R_T = \frac{R_1 \times R_2}{R_1 + R_2}$ (product $\div$ sum)

COMMON MISTAKES

- Adding parallel resistors like series (**wrong!**). Use $1/R_T$.
- Forgetting the last step: after finding $1/R_T$, **flip it**.
- Using minutes or hours in $E = Pt$ for joules. **Convert to seconds**.
- Mixing kWh with J. **Use kW $\times$ h for kWh**.
- Using mA or k Ω without converting to A or Ω .

QUICK EXAMPLE

Question: 6 Ω and 3 Ω are connected in parallel across a 12 V battery. Find R_T , the total current, the power, and the energy used in 5 minutes.

Solution:

$$R_T = \frac{6 \times 3}{6 + 3} = \frac{18}{9} = 2 \Omega$$

$$I = \frac{V}{R_T} = \frac{12}{2} = 6 \text{ A} \quad (\text{check: } 2 + 4 = 6 \text{ A})$$

$$P = IV = 6 \times 12 = 72 \text{ W}$$

$$E = Pt = 72 \times (5 \times 60) = 72 \times 300$$

Answer: $R_T = 2 \Omega$, $I = 6 \text{ A}$, $P = 72 \text{ W}$, $E = 21\,600 \text{ J}$

TRY THIS: QUICK CHECK

- A 20 Ω resistor carries 0.5 A. Find the voltage across it.
- A 60 W bulb is on for 5 hours. How many kWh does it use?
- Two 4 Ω resistors: find R_T in series and in parallel.

Answers:

- $V = IR = 0.5 \times 20 = 10 \text{ V}$
- $0.06 \times 5 = 0.3 \text{ kWh}$
- Series 8 Ω ; parallel 2 Ω